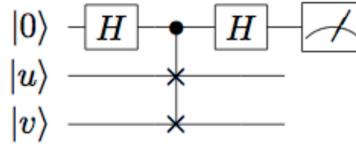


Quantum Computing Exam

8 July 2025

Problem 1. Consider the quantum circuit:



Apply the circuit to the input $|0\rangle \otimes |v\rangle \otimes |u\rangle$ with

- $|v\rangle = |u\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$.
- $|v\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $|u\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$.

What is the probability that the measurement outcome is 0 in each of the two cases?

Problem 2. Alice wants to send to Bob two bits of classical information. To do so, Alice uses the superdense encoding protocol.

Assuming that Alice and Bob can share the entangled pair

$$\frac{1}{\sqrt{2}}|01\rangle + \frac{1}{\sqrt{2}}|10\rangle,$$

which operations must perform Alice on her qubit before sending it to Bob?

Problem 3. Construct a quantum circuit implementing the two-level matrix

$$U = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & a & 0 & 0 & 0 & 0 & b & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & c & 0 & 0 & 0 & 0 & d & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

by using only $\mathbf{C}^2(\mathbf{NOT})$ and $\mathbf{C}^2(\mathbf{A})$, where $\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Write down the matrix corresponding to the $\mathbf{C}^2(\mathbf{NOT})$ gates of the resulting circuit.

Problem 4 (optional). Let $f(x): \{0, 1\} \rightarrow \{0, 1\}$ be a Boolean function. Show that $\mathbf{U}_f: |x, y\rangle \mapsto |x, y \oplus f(x)\rangle$ is unitary.

Problem 5 (optional). Si consideri lo stato

$$\frac{1}{\sqrt{2}} |00\rangle + \frac{e^{i\phi}}{\sqrt{2}} |10\rangle.$$

Possiamo stimare l'angolo ϕ applicando il Fourier sampling (cioè Hadamard su ciascun qubit seguito da una misurazione nella base standard). Supponiamo che dopo un numero sufficiente di test, il risultato 01 non venga mai ottenuto. Calcolare il valore di $\phi \in [0, 2\pi]$.